

A midpoint identity for the complementary torus order at discriminant -3

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Abstract

For an odd prime $p \neq 3$, write $\chi_3(p) = (-3/p)$ and $m_0(p) = p + \chi_3(p)$. Then

$$\frac{1}{m_0(p)} - \frac{1}{2} \left(\frac{1}{p-1} + \frac{1}{p+1} \right) = -\frac{\chi_3(p)}{p^2-1}.$$

The identity is elementary. Summing the right-hand side gives an explicit convergent series for the offset between the selected neighbour and the midpoint of $\{p-1, p+1\}$. The same character is the Kronecker symbol that distinguishes the two torus orders of Williams' $p+1$ test at $D = -3$. No asymptotic for the associated hire-count is claimed, nor a relation to $L(s, \chi_3)$ beyond the identification $\chi_3 = (-3/\cdot)$.

1 The door

Let $p > 3$ be prime and let χ_3 be the non-principal Dirichlet character modulo 3, equivalently the Kronecker symbol $(\frac{-3}{\cdot})$. Set

$$m_0(p) = p + \chi_3(p) = \begin{cases} p+1 & \text{if } p \equiv 1 \pmod{3}, \\ p-1 & \text{if } p \equiv 2 \pmod{3}. \end{cases}$$

Thus $m_0(p)$ is one of the two even neighbours of p . Williams' $p+1$ method at discriminant $D = -3$ uses the complementary order

$$m_1(p) = p - \chi_3(p).$$

The pair $\{m_0, m_1\}$ is $\{p-1, p+1\}$. The hire graph keeps m_0 and discards m_1 . Same quadratic field as the Eisenstein integers and as the CM curve $y^2 = x^3 + B$; different group.

2 The identity

Lemma 1. *For every prime $p > 3$,*

$$\frac{1}{m_0(p)} - \frac{1}{2} \left(\frac{1}{p-1} + \frac{1}{p+1} \right) = -\frac{\chi_3(p)}{p^2-1}.$$

Proof. The midpoint reciprocal is $p/(p^2-1)$. If $p \equiv 1 \pmod{3}$, then $\chi_3(p) = +1$ and $m_0(p) = p+1$, so the left-hand side is

$$\frac{1}{p+1} - \frac{p}{p^2-1} = \frac{p-1-p}{(p+1)(p-1)} = -\frac{1}{p^2-1}.$$

If $p \equiv 2 \pmod{3}$, then $\chi_3(p) = -1$ and $m_0(p) = p-1$, so the left-hand side is

$$\frac{1}{p-1} - \frac{p}{p^2-1} = \frac{p+1-p}{(p-1)(p+1)} = +\frac{1}{p^2-1}.$$

In both cases this equals $-\chi_3(p)/(p^2-1)$. □

Remark 1. *Equivalently, the midpoint is $p/(p^2-1)$ and the gap is the unit $\pm 1/(p^2-1)$. Adding them recovers $1/m_0(p)$.*

3 The offset

Summing Lemma 1 over primes $p > 3$ up to X gives

$$\sum_{5 \leq p \leq X} \frac{1}{m_0(p)} - \frac{1}{2} \sum_{5 \leq p \leq X} \left(\frac{1}{p-1} + \frac{1}{p+1} \right) = - \sum_{5 \leq p \leq X} \frac{\chi_3(p)}{p^2-1}.$$

The right-hand side converges absolutely as $X \rightarrow \infty$. Its value is

$$- \sum_{p>3} \frac{\chi_3(p)}{p^2-1} \approx 0.02598841679.$$

There is no reason to expect a form in π and $\sqrt{3}$: $L(1, \chi_3) = \pi/(3\sqrt{3})$ and $L(3, \chi_3) = 4\pi^3/(81\sqrt{3})$ are odd arguments of an odd character; the present series is a character-prime-zeta at even powers. The series is left as the series.

The unsigned neighbour sums $\sum 1/(p-1)$ and $\sum 1/(p+1)$ each diverge like $\frac{1}{2} \log \log X$. The identity says the selected door tracks the midpoint of those two diverging curves at a frozen gap. The signed series $\sum \chi_3(p)/m_0(p)$ is a different object (cancellation at $1/p$, numerical value ≈ -0.241) and is not used here.

4 Refusal

Lemma 1 does not yield an asymptotic for

$$H(X) = \#\{q \text{ prime} : 5 \leq q \leq X, \exists p \leq X, q \mid m_0(p)\}.$$

It is not an elliptic-curve method. It records one elementary relation between the complementary torus order at $D = -3$ and the two neighbours of p .